International Islamic University, Islamabad
Faculty of Engineering & Technology
Department of Electrical and Computer Engineering

CS101L

Object Oriented Programming

Lab Manual

Subject Teacher:	
Lab Instructor:	

Student Name	
Reg. No.	
Section	

Prepared by:

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International Islamic University, Islamabad
Faculty of Engineering & Technology
Department of Electrical and Computer Engineering

CS101L

Object Oriented Programming

Lab Manual

Names of Group Members

Student Name	
Reg. No.	

Student Name	
Reg. No.	

Student Name	
Reg. No.	

Student Name	
Reg. No.	

CS101L

Object Oriented Programming

Lab Manual

OBJECTIVE
The objective of this lab is to make students learn about fundamental concepts of Object-Oriented Programming and its implementation in C++ language. The lab covers the concepts of classes, objects, attributes, operator overloading, inheritance, virtual functions, and friend functions.

CLO	CLO Description	DOMAIN	PLO
01	**Develop** small to moderate programs while exhibiting the object-oriented programming skills.	C3	02
02	**Build** small software using OOP.	P2	05
03	**Participate** effectively as an individual or in a team.	A2	09

CLO: Course Learning Outcome.

PLO: Program Learning Outcome.

Object Oriented Programming Lab Rubrics			
Name:	**Reg. No.:**	**Signature:**	**Instructor:**

a) PSYCHOMOTOR (To be judged in the field/lab during experiment. check coding and take viva)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Lab	Lab 1	Lab 2	Lab 3	Lab 4	Lab 5	Lab 6	Lab 7	Lab 8	Lab 9	Lab 10	Lab 11	Lab 12	Lab 13	Lab 14
1	Program Implementation	0	1	2,3	4,5	6	Weightage	6	6	6	6	6	6	6	6	6	6	6	6	6	6
		Absent	With several critical errors	With few critical errors	With some non-critical errors	Without errors	Obtained														
	Sub Total Marks (TP)							**Obtained Marks (OP)**													

(b) AFFECTIVE (To be judged in the field/lab during experiment, Check attitude)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Lab	Lab 1	Lab 2	Lab 3	Lab 4	Lab 5	Lab 6	Lab 7	Lab 8	Lab 9	Lab 10	Lab 11	Lab 12	Lab 13	Lab 14
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	Weightage	4	4	4	4	4	4	4	4	4	4	4	4	4	4
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	Obtained														
	Sub Total Marks(TA)							**Obtained Marks (OA)**													

3	**Total Marks (TP + TA) :**		**Obtained Marks (OP + OA)**

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Object Oriented Programming Lab (CS101L)

Lab 1 : Classes and Objects

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Make a **Circle** class.

- It has three attributes **radius**, the **x** and the **y** coordinates of its center of **double** data type.
- Make **void set(double, double, double)** functions to set x, y and radius.
- Make **void get()** functions to get x, y and radius from user.
- Make **void show()** function to display x , y coordinates and radius of a circle.
- Make **double area()** function, and a **double circumference()** function to calculate and return area and circumference.
- Call these functions in **main()** to test their working.

  This should be the output of program

```
Radius: 50, x: 10, y: 20
Area          of c1 = 7854
Circumference of c1 = 314.16

Enter Radius: 10
Enter x-coordinate: 3
Enter y-coordinate: 5

Radius: 10, x: 3, y: 5
Area          of c2 = 314.16
Circumference of c2 = 62.832

Press any key to continue . . .
```

Q2. Make a **Rectangle** class.

- It has two attributes **length**, and **width** of **int** data type.
- Make two member functions **int perimeter()** and **int area()** to calculate and return **perimeter** and the **area** of a rectangle.
- Make **void set(int l, int w)** and **void get()** functions to change the **length** and **width** attributes.
- Make a **void draw()** function that draws a rectangle using a character * on console.

```
*********
*       *
*       *
*********
```

This should be the output of program

```
Length: 4
Width : 6
Area of Rectangle = 24
Perimeter of Rectangle = 20
******
*    *
*    *
******
Press any key to continue . . .
```

Solution Lab 1 Task 1:

Solution Lab 1 Task 2:

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Object Oriented Programming Lab (CS101L)

Lab 2 : Constructors

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Make a **student** class.

- It has two attributes
  - **name** as a **character array** of size **50**
  - **gpa** as a **double** data type.
- Make **no argument constructor** to set
  - **name** equal to **""** **(hint: use strcpy function)**
  - **gpa** equal to **zero**
- Make **two arguments constructor** to set
  - **name** equal to **First Argument**
  - **gpa** equal to **Second Argument**
- Make **void get()** functions to get **name** with spaces and **gpa** from user.
- Make **void show()** function to display **name** and **gpa** on screen.
- Make **three student objects** in **main()** and call two arguments constructer while creating first object and call no-argument constructer for other two objects.
- Call these functions to check their working.

This should be the output of program

```
Your Name is “Rashid Farid” and Your GPA = 2.59

Enter Name: M. Usman
Enter GPA : 3.5
Your Name is “M. Usman” and Your GPA = 3.5

Enter Name: M. Ali
Enter GPA : 4.0
Your Name is “M. Ali” and Your GPA = 4.0

Press any key to continue . . .
```

Q2. Make a **Temperature** class.

- It has one attributes **c** to represent temperature degree centigrade, of **double** data type.
- Make **no argument constructor** to set **c** equal to **zero**

- Make **void get_c()** functions to get temperature value in **Degree centigrade** from user.
- Make **void get_f()** functions to get temperature value in **Degree Fahrenheit** from user.
- Make **void show_c()** function to display temperature in **Degree Centigrade** on screen.
- Make **void show_f()** function to display temperature in **Degree Fahrenheit** on screen.

Note: ctemp = (ftemp-32) * 5.0 / 9.0;

This should be the output of program

```
Enter Temperature in Degree Centigrade: 37
37 C = 98.6 F

Enter Temperature in Degree Fahrenheit: 98.5
98.5 F = 36.9444 C
```

Solution Lab 2 Task 1:

Solution Lab 2 Task 2:

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Object Oriented Programming Lab (CS101L)

Lab 3 : Destructors

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Make a **Tree** class.

- It has one attributes
  - **height** as **unsigned int**
- Make **No Argument Constructor** to set **height** equals to the **zero.** Also print this message **"A tree of zero length has been created".**
- Make **One Argument Constructor** to set **height** equals to the argument value. Also print this message **"A tree has been created".**
- Make a **destructor** and print **"The tree has been destroyed"** message inside it.
- Make **void grow(int meters)** function to add argument value to the tree **height** value.
- Make **void printsize()** function to print tree **height** on screen.
- **Make void Create_Tree()** function outside the class.
- Make **One tree object** in **Create_Tree()** function **and call members functions inside it.**
- **Call Create_Tree() function two times inside main().**

The output of program should be like this.

```
A tree has been created
Tree height is 16 Meters
The Tree has been destroyed
A tree has been created
Tree height is 16 Meters
The Tree has been destroyed
Press any key to continue . . .
```

Q2. Make an **Employee** class.

- It has two attributes
  - **id** as **unsigned int.**
  - **length** as **unsigned int.**
  - **name** as a **character pointer.**
- Make **no argument constructor** to set
  - **ID** equals to 1
  - **length** equals to 100

- dynamically allocate length+1 = 101 bytes of memory to name pointer

- Make **two arguments constructor** to set
  - **ID** equals to first argument
  - **length** equals to string length of 2nd argument
  - **name** equals to the copy of second argument dynamically .
- Make **a destructor** and free the dynamic allocated memory.
- Make **void get()** functions to get ID and **name** with spaces from user. **(hint: use cin.get or cin.getline function)**
  - **Empty input buffer using fflush(stdin) after using cin.get**
- Make **void show()** function to display ID and **name** on screen.
- Make **one Employee Object** in **main()** using no argument constructor and another **Employee Object** using two argument constructor.
- Call functions to check their working.

Following is the possible output of program.

```
ID = 1, Name = M. Usman

Enter Employee ID: 1
Enter Employee Name: Rashid Farid

ID = 1, Name = Rashid F

Enter Employee ID: 2
Enter Employee Name: Shahid Farid Chishti
ID = 2, Name = Shahid Farid Chishti
Press any key to continue . . .
```

Solution Lab 3 Task 1:

Solution Lab 3 Task 2:

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Object Oriented Programming Lab (CS101L)

Lab 4 : Passing Objects as Function Arguments.

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Make a **Matrix** class, which represents a 3x3 Matrix

- It has three attribute
  - **ary** as a ThwoD **array** of **double** data type.
  - **rows** as unsigned int.
  - **columns** as unsigned int.
- Make **no argument constructor** to set
  - **All elements** of **ary** equal to **zero.**
  - Set **rows** and **columns equal to 3.**
- Make **three arguments constructor** that **takes**
  - first argument as **TwoD array of double** data type
  - 2nd argument as number of **rows**
  - 3rd argument as number of **columns**
- Make **void get()** functions to get **data** for each element of array **ary.**
- Make **void show()** function to display **ary** on screen.
- Make **double mod()** function to calculate and return mod value of the matrix.
- Make **void add(Matrix A, Matrix B) function to add two Matrix Objects and copy the answer in the third object which has called the add function. e.g. Obj3.add(Obj1 , Obj2);**

OR

- you can make **Matrix add(Matrix B)** function to add two Matrix Objects and return the answer as a Matrix object.

Similarly

- Make **void Sub(Matrix A, Matrix B)** to subtract two Matrix object
- Make **void Mul(Matrix A, Matrix B)** to Multiply two Mat object
- Remember to write functions **declaration** inside the class and functions **definition** outside the class.
- **Make three Matrix objects in main()** and call these functions to check their working.

Here is the output of program

```
A[0][0]:1
A[0][1]:2
A[0][2]:3
A[1][0]:4
A[1][1]:5
A[1][2]:6
A[2][0]:7
A[2][1]:8
A[2][2]:9
   9   8   7
   6   5   4
   3   2   1
 *
   1   2   3
   4   5   6
   7   8   9
 =
  90 114 138
  54  69  84
  18  24  30
Press any key to continue . . .
```

Solution Lab 4 Task:

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Object Oriented Programming Lab (CS101L)

Lab 5 : Passing Objects by Reference.

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Make a **Vector** class that represents a 3D vector

- It has three attribute
  - **i, j and k** as double data type.
- Make **no argument constructor** to set
  - **All attributes** equal to **zero.**
- Make **three arguments constructor** that **takes**
  - First argument and saves it into **i.**
  - Second argument and saves it into **j.**
  - Third argument and saves it into **k.**
- Make **void Get()** functions to get **data** for attributes of this class.
- Make **void Show()** function to display these **attributes** on screen. Like this: **< 3, 4, -6 >**
- Make **double Mod()** function to calculate and return length of a vector.

$$|A| = \sqrt{A_i^2 + A_j^2 + A_k^2}$$

- Make **double Dot(const Vector& B) const** function to calculate and return dot product of two vectors.

$$|A \cdot B| = A_i B_i + A_j B_j + A_k B_k$$

- Make **Vector Cross(const Vector& B) const function to calculate and return cross product of two vectors. This function will be called in main() like this:**

```
C = A.Cross(Vector B);
```

$$|A \times B| = \begin{vmatrix} i & j & k \\ A_i & A_j & A_k \\ B_i & B_j & B_k \end{vmatrix}$$

Similarly

- Make **Vector Sub(const Vector& B) const** to subtract two Vector objects
- Make **Vector Add(const Vector& B) const** to add two Vector objects
- Remember to write functions **declaration** inside the class and functions **definition** outside the class.
- Also make three separate files **Vector.h**, **Vector.cpp** and **main.cpp**
- **Make three Vector objects in main()** and call these functions to check their working.
    - In the main function take inputs for two vectors from user.
    - Ask the user what type of operation he/she wants to perform.
    - Apply that operation and show the answer on screen.

Here is the output of program

```
Enter Vector A:
Enter Coefficient of <i>:1
Enter Coefficient of <j>:2
Enter Coefficient of <k>:3
Enter Vector B:
Enter Coefficient of <i>:4
Enter Coefficient of <j>:5
Enter Coefficient of <k>:6
Enter Vector Operation:
[1] - Magnitude of A Vector.
[2] - Dot Product.
[3] - Cross Product.
[4] - A - B
[5] - A + B
[q] - Quit this Program
3
< 1, 2, 3 > Cross < 4, 5, 6 > = < -3, 6, -3 >
Press any key to continue . . .
```

Solution Lab 5 Task:

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Object Oriented Programming Lab (CS101L)

Lab 6 : Unary Operator Overloading

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Create a class called **Time** that represents a time in 24 hours format.

- It has three attribute
  - **sec**, **min** and **hrs** as integer data type.
- Make no argument constructor to set initial time equals to **00:00:00** as **hh : mm : ss**
- Make three argument constructor to set initial time equals to the argument values.
  - First argument changes **hrs**.
  - Second argument changes **min**.
  - Third argument chnges **sec**.
- Make **void set(int h,int m,int s)** function to set **hrs**, **min**, **sec**.
- Make **void show()** function to display time in **23:59:59** format.
- Make **void get()** function to get values of **hrs**, **min** and **sec** from user.
- Make **void tick()** member function that increments the time stored in a Time object by one second.
- Make **Time Operator ++()** member function that increments the time stored in a Time object by one second in a **prefix** way.
- Make **Time Operator ++(int)** member function that increments the time stored in a Time object by one second in a **postfix** way.
- Make **Time Operator --()** member function that decrements the time stored in a Time object by one second in a **prefix** way.
- Make **Time Operator --(int)** member function that decrements the time stored in a Time object by one second in a **postfix** way.
  - While incrementing or decrementing time make sure to implement the following cases:
    - Incrementing into the next minute.
    - Incrementing into the next hour.
    - Incrementing into the next day (i.e., 23:59:59 to 00:00:00).
- In the main function.

- Make a time object in a main function.
- Get the initial time value from user.
- Make an infinite loop
- Call **tick()** and **show()** functions in that loop for time object after each second.

- You can use **windows.h** header file and **Sleep(1000)** function to get delay of 1 second.

Here is the output of program

```
Enter Hours  :23
Enter mmutes:59
Enter ssonds:55
23:59:56
23:59:57
23:59:58
23:59:59
00:00:00
00:00:01
00:00:02
00:00:03
00:00:04
00:00:05
00:00:06
00:00:07
00:00:08
00:00:09
00:00:10
00:00:11
```

Solution Lab 6 Task:

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Object Oriented Programming Lab (CS101L)

Lab 7 : Binary Operator Overloading

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1. Create a class called **Rational** that represents a rational number.

- It has two attribute
  - **P**, and **Q** as integer data type.
- Make no argument constructor to set initial value equals to **1.**
- Make two argument constructor to set initial value equal to the argument value.
  - First argument changes **P**.
  - Second argument changes **Q**.
- Make **void Set(int p, int q)** function to set **P** and **Q**.
- Make **void Get()** function to get values of **P** and **Q** from user.
- Make **void Show()** function to display number in **P / Q** format.
- Make **int GCD()** function to calculate and return GCD of **P** and **Q**.
- Make **void Reduce()** member function that stores **P** and **Q** in reduced form. For example, the fraction **2/4** would be stored in the object as **1** in the numerator and **2** in the denominator. So to get **P** and **Q** in reduced form, divide both **P** and **Q** with their GCD.
- Make **double Return_Double()** member function that divides **p** by **q** and returns the answer as **double** data type.
- Make **Rational Operator + (Rational B)** member function that adds two rational numbers.
- Make **Rational Operator - (Rational B)** member function that subtracts two rational numbers.
- Make **Rational Operator * (Rational B)** member function that multiplies two rational numbers in reduced form.
- Make **Rational Operator / (Rational B)** member function that divides two rational numbers in reduced form.
  - While Performing Arithmetic Operations (+, -, * and /) on two rational numbers, make sure to return the final answer in reduced form.
- Make three rational object in a main function and call these operators to check their functionality.

Here is the output of program

```
Enter 1st Rational Number:
Enter P : 45
Enter Q : 20
Enter 2nd Rational Number:
Enter P : 21
Enter Q : 70
9/4 + 3/10 = 51/20
9/4 - 3/10 = 39/20
9/4 * 3/10 = 27/40
9/4 / 3/10 = 15/2
Press any key to continue . . .
```

Solution Lab 7 Task:

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Object Oriented Programming Lab (CS101L)

Lab 8 : Inheritance

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

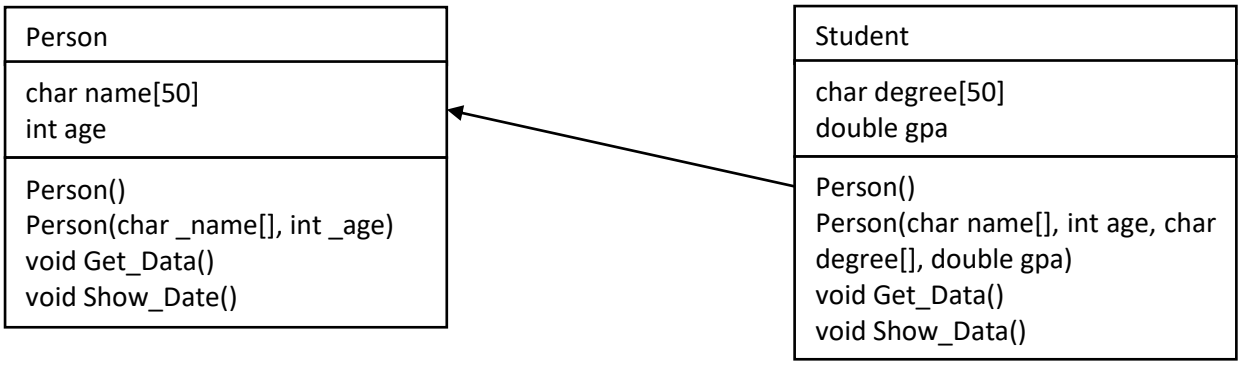


Figure: UML Diagram of inheritance

Q1A. Create a class named **Person.**

- It has two attribute as **private**
  - **name** as character array of size 50.
  - **age** as int.
- Make **no argument constructor** to set **name** equals to empty string and **age** equals to zero.
- Make **two argument constructor**
  - The first argument changes the value of **name.**
  - The second argument changes the value of **age.**
- Make **void Get_Data()** function to get **name** and **age** from user.
- Make **void Show_Data()** function show the value of **name** and **age** on screen.

Q1B. Create another class named **Student** that is publically inherited from **Person** class.

- It has two attribute as **private**
  - **Degree** as character array of size 50;
  - **gpa** as double data type.
- Make **no argument constructor** to set **degree** equals to empty string and **gpa** equals to **0.**
  - Also call the no argument constructor of parent class.
- Make **four argument constructor.**
  - The first argument changes the value of **name.**

- The second argument changes the value of **age.**
- The third argument changes the value of **degree.**
- The fourth argument changes the value of **gpa.**
- To change **name** and **age,** call two argument constructor of parent class.

- Make **void Get_Data()** function to get value of **name, age, degree** and **gpa** from user.
  - Also call **Get_Data()** function of parent class from child class.
- Make **void Show_Data()** function to show value of **name, age, degree** and **gpa** on screen.
  - Also call **Show_Data()** function of parent class from child class.

Q1C. Make main() function

- Make a student object. Get and then show his/her information on screen.

Here is the output of program

```
Enter Student's Information

Name: Rashid Farid Chishti
Age : 42
Degree Program: BS in Computer Engineering
GPA : 3.59

[You typed Following Information]

Name          : Rashid Farid Chishti
Age           : 42
Degree Program: BS in Computer Engineering
GPA           : 3.59
Press any key to continue . . .
```

Solution Lab 8 Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 9 : Using Inheritance in Date and DoB

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

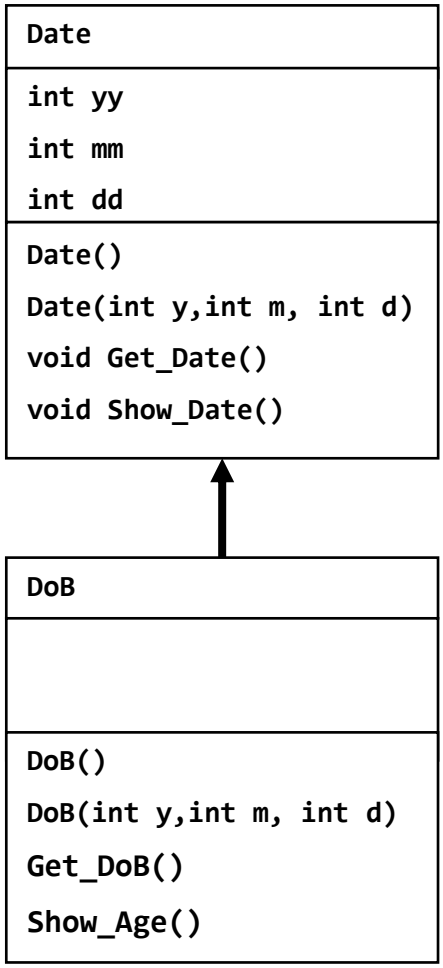
(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Algorithm Design or Data Record, Analysis and Evaluation	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Q1A. Create a class named **Date.**

- It has three **public** attributes:
  - **yy** as integer
  - **mm** as integer
  - **dd** as integer
- Make **no argument constructor** to set yy, mm and dd equals to current date.
- Make **three arguments constructor.**
- Make a **void Get_Date()** function to get data from user and this is a sample output of a function:
  - Enter Year [2014]: 2023
  - Enter Month[1-12]: 11
  - Enter Day [1-31]: 6
- Make **void Show_Date()** function to show date like this format
  Current Date = November 6, 2023
- For this purpose, make an array of month names at the top of code.



Q1B. Create another class named **Date_of_Birth** that is publically inherited from **Date** class.

- Make **no argument** and **three arguments constructors.**
- Make a **void Get_DoB()** function that gets user's Date of birth like this:
  Enter Year [2014]: 1998
  Enter Month[1-12]: 12
  Enter Day [1-31]: 25
- Make a **void Show_DoB()** function that shows user Date of Birth like this format.
  Date of Birth = December 25, 1998

- Make a **void Show_Age()** function that calculates and then shows user age based on current date and date of birth, like this format.

  Age = 24 years and 11 months.

  - While subtracting months, take one borrow from year if needed.

Q1C. Make main() function

- Make an Object of **Date** class and show current Date on screen.
- Make an Object of **Date_of_Birth** class. Get user's date of birth and show his/her age on screen..

This is a possible output of the program

```
Current Date = November 6, 2023

Please Enter Your Date of Birth
Enter Year [2014]: 1998
Enter Month[1-12]: 12
Enter Day  [1-31]: 25

Date of Birth = December 25, 1998
Age = 24 years, 11 months
Press any key to continue . . .
```

Solution Lab 9 Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 10 : Multiple Inheritance

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Algorithm Design or Data Record, Analysis and Evaluation	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

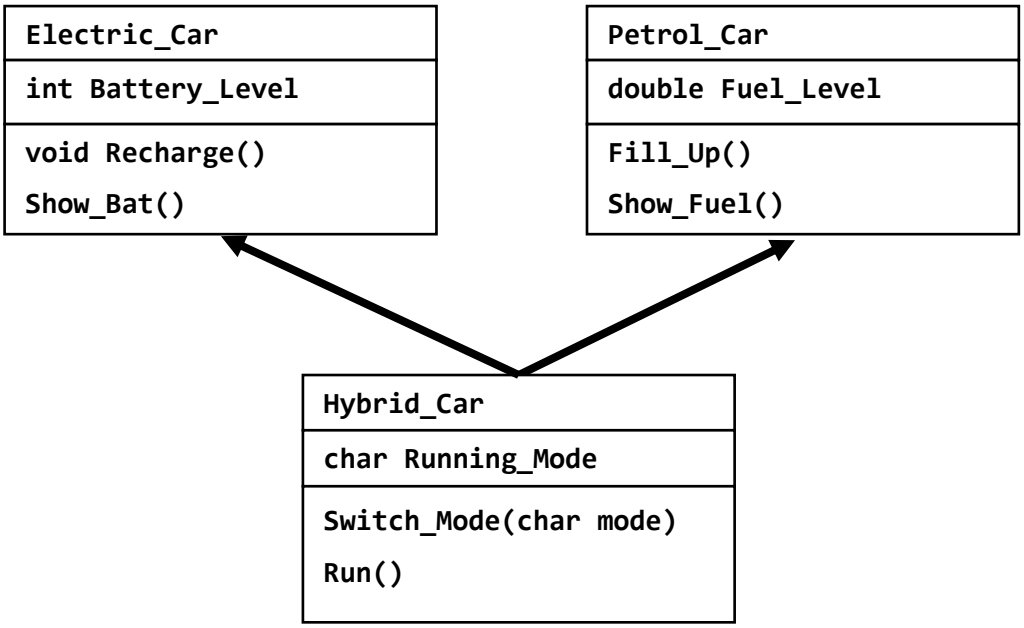


Figure: UML Diagram of Multiple Inheritance

Q1A. Create a class **Electric_Car**

- It has an attribute **Battery_Level** (type **int**) for a value from **0** to **100 %**
- Make **Recharge**() function that adds a number **1** to the **Battery_Level.**
- Make **Show_Bat**() function that shows current **Battery_Level** on screen.

Q1B. Create a class **Petrol_Car**

- It has an attribute **Fuel_Level** (type **double**) for a value from **0** to **50 Liter**
- Make **Fill_Up**() function that adds a number **0.5** to the **FuelLevel.**
- Make **Show_Fuel**() function that shows current **Fuel_Level** on screen.

Q1C. Create a class **Hybrid_Car**

- It has an attribute **Running_Mode** (type **chr**).
  - If **Running_Mode** is b then car is on Battery mode.
  - If **Running_Mode** is f then car is on Fuel mode.
- Make **Switch_Mode(char mode)** function that changes the **RunninMode** to **b** or **f** based on function arguments.
- Make **Run()** function that
  - subtracts 1 from **Battery_Level** if car is on Battery mode
  - subtracts 0.5 from **Fuel_Level** if car is on Fuel mode.

Q1D. Make an Object of **Hybrid_Car** in **main()** function and show the simulation results on screen.

- Generate random numbers from 0 to 5
  - If number equals to 0, call Fill_Up()
  - If number equals to 1, call Recharge()
  - If number equals to 2, call Switch_Mode('b')
  - If number equals to 3, call Switch_Mode('f')
  - If number equals to 4, call Run()
  - If number equals to 5, call Run()

This is the output format of Simulation

```
Battery Level = 18 %
Fuel    Level = 2.0 Liter
```

Solution Lab Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 11 : Virtual Functions

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1.5	3	4.5	6	
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

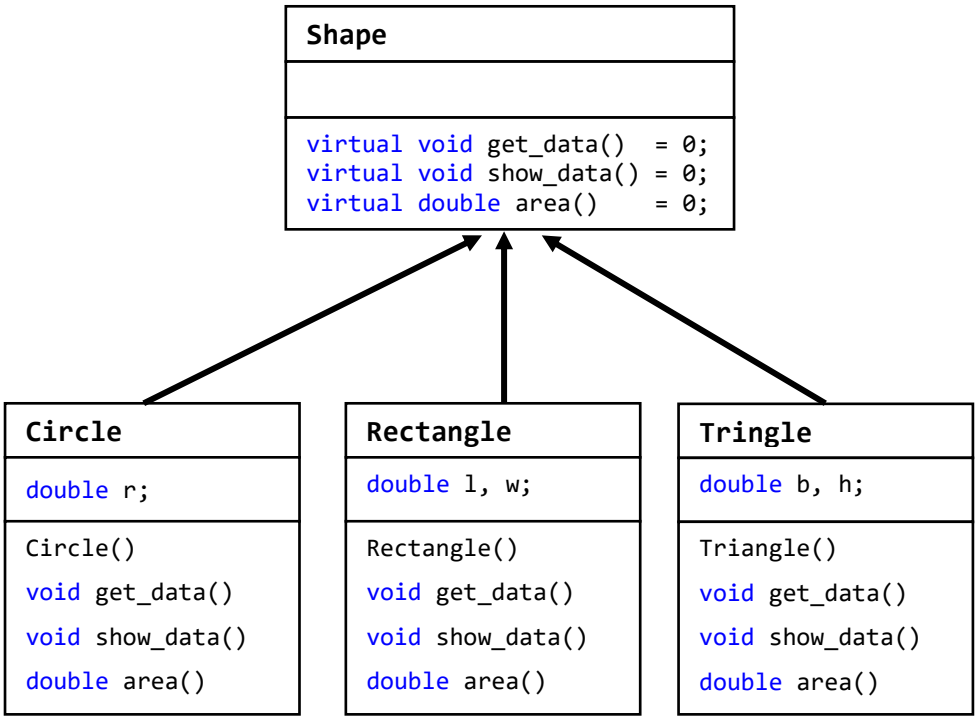


Figure: UML Diagram of Virtual Functions

- Create a class **Shape** that has three pure virtual functions as shown in the UML Diagram above.
- From this class derive three classes named **Circle**, **Rectangle** and **Triangle**.
    - Each of these three classes should have a **get_data**() function to get its data from the user using keyboard, a **show_data**() function to display its data and an **area**() function to calculate and return area of that class.
- Write a **main()** program that creates an array of 100 pointers to **Shape**.
    -
    -
    - Aske User: How many Shapes you want to enter?

- In a loop, ask the user which object he wants to enter (c = circle, r = rectange, t = triange: c)
- Using **new** operator create an object of user choice to hold data and put the pointer to that object in the array.
- When the user has finished entering the data for all objects, display the resulting data and the area of that object using a **for** loop and a single statement such as

  All_Shapes[j]-> show_data();

This is the output of Program on screen.

```
How many Shapes you want to enter? 4
[Options] c = circle, r = rectange, t = triange: a
[Options] c = circle, r = rectange, t = triange: c
Enter Radius: 1
[Options] c = circle, r = rectange, t = triange: t
Enter Base   : 2
Enter Height : 3
[Options] c = circle, r = rectange, t = triange: r
Enter Length: 4
Enter Width : 5
[Options] c = circle, r = rectange, t = triange: c
Enter Radius: 6
[0]->[Circle   ] Radius = 1 ,                Area = 3.1416
[1]->[Triangle ] Base   = 2 , Height = 3 ,   Area = 3
[2]->[Rectangle] Length = 4 , Width  = 5 ,   Area = 20
[3]->[Circle   ] Radius = 6 ,                Area = 113.098
Press any key to continue . . .
```

Solution Lab Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 12 : Friend Functions

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

- Create a class **Length** that has two attributes
  - **Feet** as **integer** data type
  - **Inches** as **float** data type
- Make a **no argument constructor** to set **Feet** and **Inches** equals to **zero.**
- Make a **one argument constructor** that takes length in the form of float feet and then it separates feet and inches and stores them in the **Feet** and **Inches** attributes. For example, if input argument is **2.5** then the value of **Feet** will be **2** and the value of **Inches** will be **6.0.**
- Make a **two argument constructor** that takes **feet** in first argument and **inches** in the second argument and stores them in the respective attributes.
- Make a **Show()** member function to show length in the form **2'-6.0"** where **2** is the value of **Feet** and **6.0** is the value of **Inches.**
- Declare a *** operator** as **friend** in the **Length** class and this Operator takes two **Length** objects as arguments.
- Define *** operator** outside the class to Multiply two Length Objects.
- Hint:
  - Fist convert two length objects into **float feet.**
  - Multiply two **float feet** numbers.
  - Convert the answer back to **Feet** and **Inches** using **one argument constructor** that you have already defined.
  - Return the answer in the form of **Length** object.

- For example,
  - I have two lengths **2'-6.0"** and **3'-3.0".**
  - Converting them to float feet I have **2.5** float feet and **3.25** float feet.
  - Now multiply both and I got 2.5 * 3.25 = **8.125** float feet.
  - Converting **8.125** float feet to feet and inches. I got **8 feet** and 0.125*12 = **1.5 inches.**

- Make **int main()** function to test **friend operator *.**

Solution Lab Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 13 : Static Functions

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Make an **Employee** class.

- It has two attributes
  - **id** as **unsigned int**
  - **name** as a character array of size 100.
  - **total** as **static integer** to show total number of employees
  - Set the initial value of **total** equals to 0.
- Make **no argument constructor**
  - Increment **total.**
  - Set **id** equals to **total** and **name** equals to **empty sting.**
- Make a **destructor** and decrement **total** inside the destructor. Also, show **id** and **name** of the Employee object that is being destroyed.
- Make **void get_data()** functions to get **name** with spaces from user. **Hint**: use cin.getline(array_name, size).
- Make **void show_data()** function to display **id** and **name** on screen.
- Make **static void show_total()** to show **total** number of employees.
- Make main() function
  - Call **show_total()** using class name.
  - Make an Employee Object by name and get data from user using keyboard.
  - Call **show_total()** using class name.
  - Now Make an Employee Object Dynamically using Employee pointer.
  - Call **show_total()**
  - Destroy the this employee object.
  - Again call **show_total()** using class name.

This is the output of Program on screen.

```
Total Employees are: 0
Enter Name: Rashid Farid

Your ID = 1 and Your Name = Rashid Farid

Total Employees are: 1
Enter Name: Muhammad Ibrahim

Your ID = 2 and Your Name = Muhammad Ibrahim
Destroying ID: 2 of Employee: Muhammad Ibrahim

Total Employees are: 1
Press any key to continue . . .
```

Solution Lab Task:

International Islamic University Islamabad
Faculty of Engineering and Technology
Department of Electrical Engineering

Object Oriented Programming Lab (CS101L)

Lab 14 : File Handling in C++

Name:	
Reg. No:	
Date of Experiment:	

OBE Rubrics Evaluation

a) PSYCHOMOTOR (To be judged in the field/lab during experiment)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
1	Practical Implementation	0	1	2,3	4,5	6	1
		Absent	With several critical errors and incomplete	With few errors, and incomplete	With some errors and complete	Without errors and complete	

(b) AFFECTIVE (To be judged on the copy of experiment submitted)

Sr. No.	Criteria	Level 1 (0%)	Level 2 (25%)	Level 3 (50%)	Level 4 (75%)	Level 5 (100%)	Marks Obtained
2	Level of Participation & Attitude to Achieve Individual/Group Goals	0	1	2	3	4	
		Absent	Bad Attitude	Decent Attitude	Good Attitude	Proactive Attitude	

3	**Total Marks Obtained (Out of 10):**	

Make an **Employee** class.

- It has two attributes
  - **Name** as **character array**
  - **Salary** as **an integer.**
- Make **Add()** function that takes **Name** and **Salary** and store in File named **"employees.xt"**
- Make **Show_All()** function that shows **Names** of All Employees and their **Salary** on screen by reading data from "employees.xt" file.
- Make **Search()** function that searches **Name** of Employee on screen by reading data from "employees.xt" file.
- Note: Use Formatted File I/O. Inside "employees.xt" file one line should contain data of one employee only.

This is the output of program

```
[Operations]
A - Add
R - Read All
S - Search name
Q - Quit
a
Enter name  : M. Ibrahim
Enter Salary  : 50345
Press any key to continue . . .
```

```
[Operations]
A - Add
R - Read All
S - Search name
Q - Quit
a
Enter name  : M. Bilal
Enter Salary  : 67345
Press any key to continue . . .
```

```
[Operations]
A - Add
R - Read All
S - Search name
Q - Quit
r
#1
  name  = M. Ibrahim
  salary  = 50345
#2
  name  = M. Bilal
  salary  = 67345
```

```
[Operations]
A - Add
R - Read All
S - Search name
Q - Quit
s
Enter name  to Search: lal
#2
name  = M. Bilal
Salary  = 67345
```

Solution Lab Task: